CausalMedFusion: A Multimodal Causal Framework for Chest Radiology Analysis

Mini project report submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science & Design

Submitted by

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CERTIFICATE

This is to certify that the Mini project report for the project entitled "**CausalMedFusion: A Multimodal Causal Framework for Chest Radiology Analysis**" is a bonafide report of the project presented during VIth semester (CXD334 - Mini Project) by **Anuroop P V(FIT23CSD015)**, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B.Tech) in Computer Science & Design during the academic year 2025-2026.

Dr. Paul P Mathai

Project Coordinator | Project Guide | Head of the Department

ABSTRACT

The integration of heterogeneous clinical data for predictive modeling in intensive care units (ICUs) remains a critical challenge in clinical artificial intelligence. Existing approaches typically process imaging, clinical text, laboratory measurements, and vital signs independently, overlooking their temporal interdependencies during patient progression.

This project introduces CausalMedFusion, a multimodal learning framework designed to model the temporal and causal evolution of patient state within the first 24 hours of ICU admission. The system structures raw multimodal inputs—including chest X-rays, clinical notes, laboratory data, and vital signs—into sequential time windows, enabling unified representation learning across modalities and time.

At the core of the framework is a Transformer-based architecture that performs intra-window multimodal fusion and inter-window temporal reasoning through causally constrained attention mechanisms. A two-stage training strategy is employed: first, the model learns clinically meaningful severity representations using a hybrid objective combining next-window prediction, intervention-aware supervision, and temporal discrimination; second, these representations are leveraged for downstream mortality prediction.

Preliminary results demonstrate that modeling temporally aligned multimodal interactions significantly improves the quality of learned patient representations and enhances predictive performance compared to modality-independent baselines. The proposed causal severity modeling framework provides improved interpretability by aligning intermediate representations with clinically meaningful progression.

This work contributes a novel approach to multimodal clinical modeling by explicitly incorporating temporal causality into representation learning. The resulting system has the potential to support early risk stratification and informed clinical decision-making, advancing the development of robust, real-time ICU decision support systems.

Contribution by Author

Author Contribution:

- Full frontend implementation: patient registration, visit management, multimodal assessment upload, and results visualisation.
- System architecture design: full-stack integration of React frontend, Django REST API, Celery/Redis task queue, FastAPI microservices, HDF5 and PostgreSQL storage layers.
- Data pipeline development: four independent microservices for CXR embedding, report embedding, vitals processing (30-minute binning, z-score normalisation), and lab measurement extraction, all processing MIMIC-IV, MIMIC-CXR-JPG, and MIMIC-IV-Note data.
- Deep learning model design and implementation: WindowEncoder (multi-head Transformer), SeverityEncoder (causal block cross-attention), SeverityPooling, and output heads for mortality and intervention prediction.
- Two-stage training loop: Stage 1 hybrid loss with EMA momentum encoder and window order discriminator; Stage 2 mortality fine-tuning with frozen severity representations.
- Model export: TorchScript tracing, ONNX export, and per-window isotonic calibration.

Anuroop P V

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Chapter 1

Introduction

1.1 Overview

Intensive care units (ICUs) generate diverse and continuous clinical data, including chest radiographs, clinical notes, vital signs, and laboratory measurements. Each modality reflects a different aspect of patient physiology, yet effective clinical decision-making depends on their joint interpretation over time. Clinicians naturally integrate these signals to assess patient condition, but most existing artificial intelligence models fail to replicate this process.

Current approaches are largely unimodal or rely on late fusion of independently learned representations, limiting their ability to capture cross-modal interactions. Furthermore, many models operate on static snapshots rather than longitudinal patient trajectories, ignoring the temporal evolution that is critical in ICU prognostication. This motivates the central research question: how can multimodal clinical data be integrated in a temporally consistent and causally meaningful manner to improve predictive modeling in critical care?

CausalMedFusion addresses this challenge by modeling ICU data as a causal temporal process. A 24-hour stay is divided into six four-hour windows, within which multimodal inputs—imaging, text, vitals, and laboratory data—are tokenized and jointly processed using a Transformer-based encoder. Temporal dependencies are captured through a SeverityEncoder that attends only to past windows, enforcing causal consistency and preventing information leakage. The resulting severity representations are used to predict clinical interventions and in-hospital mortality through a two-stage training framework.

The system is developed using large-scale clinical datasets, including MIMIC-IV for structured electronic health records, MIMIC-CXR-JPG for imaging data, and MIMIC-IV-Note for clinical text. By large datasets and widely accessible machine learning frameworks, the project addresses a critical need for reproducible and extensible clinical AI research.

The social relevance of this work is significant. ICU environments, particularly in resource-constrained settings, face high patient loads and limited clinical capacity. Delays in recognizing patient deterioration can lead to severe outcomes. A system capable of synthesizing multimodal data into an interpretable, time-resolved risk profile can support early intervention, improve resource allocation, and reduce mortality.

The framework has broader applications beyond mortality prediction, including early warning systems, treatment decision support, patient trajectory analysis, and retrospective clinical research. Its modular design allows integration into real-time monitoring systems and hospital workflows.

However, important ethical and professional considerations must be addressed. Ensuring patient data privacy is critical when handling sensitive multimodal information. Bias in training data may lead to inequitable predictions, requiring careful evaluation and mitigation. Transparency and interpretability are essential for clinical adoption, as practitioners must trust and understand model outputs. These systems should function as decision-support tools, complementing rather than replacing clinical expertise.

In summary, this project introduces a causally grounded, temporally aware multimodal framework for ICU prediction. By aligning machine learning models more closely with clinical reasoning processes, CausalMedFusion contributes to the development of more robust, interpretable, and impactful decision support systems in critical care.

1.2 Problem Statement

Intensive care units (ICUs) operate in high-stakes environments where rapid clinical decisions must be made based on diverse and continuously evolving patient data, including imaging, laboratory results, vital signs, and clinical notes. Despite the availability of such rich multimodal information, existing AI-based clinical decision support systems are largely unimodal and fail to capture the temporal progression of patient deterioration, limiting their effectiveness.

This research addresses the need for a unified framework that can integrate multimodal clinical data over time in a causally consistent manner. The central problem is how to model a patient's condition during the first 24 hours of ICU admission to generate reliable, time-resolved predictions of severity, intervention

requirements, and mortality risk.

This limitation affects a broad population, as ICU patients worldwide are subject to delayed or suboptimal decision-making due to cognitive overload and fragmented data interpretation. Specifically, in datasets such as MIMIC-IV, irregular sampling, missing observations, and class imbalance further complicate predictive modeling.

Addressing this problem can enable timely, data-driven clinical insights and improve the reliability of AI-assisted decision-making in critical care.

1.3 Objectives

- To develop a robust multimodal data pipeline for processing and integrating chest X-ray images, clinical text, vital signs, and laboratory measurements from publicly available ICU datasets.
- To model the temporal progression of patient state by structuring a 24-hour ICU stay into sequential time windows capturing multimodal observations.
- To design a Transformer-based architecture for effective fusion of heterogeneous clinical modalities within each time window.
- To incorporate causal temporal modeling by learning severity representations that respect the chronological flow of clinical data.
- To train the proposed framework using a multi-stage learning strategy for accurate prediction of patient severity, clinical interventions, and mortality risk.
- To implement an end-to-end system capable of generating interpretable, time-resolved predictions for clinical decision support.
- To ensure the deployability and scalability of the system for real-world clinical environments.

1.4 Scope of the project

The scope of this project covers the end-to-end development of a multimodal clinical AI system, from data processing to system deployment. The work focuses on modeling patient data from the first 24 hours of an ICU stay, structured into six sequential four-hour windows. The system integrates multiple data modalities,

including chest X-ray images, clinical text (radiology and physician notes), vital sign time-series, and laboratory measurements.

The model is designed to generate window-level predictions for key clinical outcomes, including mechanical ventilation, dialysis, cardiac and mechanical support, overall patient severity, and in-hospital mortality. These predictions are expressed as probabilistic estimates to support time-resolved clinical assessment.

The project is limited to experimentation on publicly available datasets, specifically the MIMIC family of datasets, ensuring reproducibility and accessibility. It does not include real-time hospital integration or direct clinical validation. The developed web application serves as a prototype to demonstrate the complete pipeline rather than a certified clinical decision-making tool.

1.5 Social Relevance

CausalMedFusion contributes to SDG 3 (Good Health and Well-being) by enabling earlier identification of patient deterioration in intensive care units, which can support timely clinical interventions and potentially reduce preventable mortality. By integrating diverse clinical data into a unified, interpretable framework, the system helps alleviate clinician cognitive burden and improves decision-making in high-pressure environments.

The project also promotes equitable access to healthcare innovation by leveraging publicly available clinical datasets, ensuring that the research is reproducible and accessible to institutions worldwide, including those in resource-constrained settings.

In addition, CausalMedFusion aligns with SDG 9 (Industry, Innovation and Infrastructure) by demonstrating a scalable and modular architecture for multimodal clinical AI systems. Its design allows adaptation to different healthcare infrastructures, facilitating future integration into hospital workflows and encouraging further innovation in data-driven clinical decision support.

1.6 Organization of the report

This mini project report is organized into seven chapters. Chapter 1 introduces the problem statement, objectives, scope, and social relevance of the project. Chapter 2 presents a review of existing literature on multimodal clinical AI, ICU prediction,

and relevant temporal and self-supervised learning techniques.

Chapter 3 describes the system design, including software requirements, overall architecture, API design, database structure, and the technology stack. Chapter 4 details the implementation of the system, covering the data processing pipeline, microservices, and the deep learning model along with its training procedure.

Chapter 5 outlines the testing strategy, including unit testing of individual components, integration testing, and end-to-end system validation. Chapter 6 discusses the experimental results, compares the proposed system with existing approaches, and evaluates its broader impact. Finally, Chapter 7 concludes the report and highlights potential directions for future work.

Chapter 2

Literature Review

Following the introduction of the project and its significance, this chapter reviews the key prior works that inform and motivate the design of CausalMedFusion. Existing research in clinical artificial intelligence highlights both the potential and limitations of current approaches in handling complex ICU data.

The review is organized around three main areas relevant to this work: multimodal clinical data and benchmark datasets, temporal and Transformer-based modeling techniques for clinical time-series, and self-supervised and contrastive learning methods used for representation learning. These areas collectively provide the foundation for understanding the challenges addressed by this project and the advancements it seeks to contribute.

2.1 Related work

2.1.1 Clinical Datasets and Benchmarks

The MIMIC-IV dataset [1,2] is a large, publicly available electronic health record (EHR) database containing detailed information on ICU patients, including diagnoses, laboratory results, procedures, and vital signs. It forms the primary source of structured clinical data used in this work. The MIMIC-CXR-JPG dataset [3,4] extends this resource with over 377,000 chest radiographs linked to corresponding admissions, providing both image data and structured labels. Additionally, MIMIC-IV-Note [5] offers de-identified free-text clinical notes, including radiology reports and discharge summaries. These datasets are distributed through PhysioNet [6], enabling reproducible clinical AI research.

Irvin et al. [12] introduced the CheXpert dataset, demonstrating that large-scale weakly supervised learning can achieve near radiologist-level performance in chest X-ray classification. Their uncertainty-aware labeling approach informs the handling of ambiguity in imaging data.

2.1.2 Multimodal Clinical AI

Soenksen et al. [8] proposed HAIM (Holistic AI in Medicine), a multimodal framework combining tabular, time-series, text, and imaging data for ICU prediction tasks. Their work demonstrated that multimodal learning improves performance over unimodal baselines, although it relies on late fusion of independently trained representations.

CausalMedFusion extends this approach by enabling early fusion at the token level within a shared Transformer architecture, allowing richer cross-modal interactions. Furthermore, unlike HAIM, it incorporates explicit temporal causality constraints to model patient progression.

Alsentzer et al. [7] introduced BioClinicalBERT, a domain-specific language model trained on clinical notes. It has become a standard for clinical text representation and is used in this work for encoding free-text inputs.

2.1.3 Temporal and Transformer-based Modelling

Kazemi et al. [9] proposed Time2Vec, a learnable time encoding method that captures both linear and periodic temporal patterns. It is particularly suitable for irregular clinical time-series and is used in this project to encode temporal information within and across time windows.

Lim et al. [10] introduced the Temporal Fusion Transformer (TFT), a model designed for interpretable multi-horizon time-series forecasting. Its use of attention mechanisms and temporal feature selection informs the design of the temporal reasoning components in this work, particularly in learning meaningful representations from sequential clinical data.

2.1.4 Self-supervised and Representation Learning

Grill et al. [11] proposed Bootstrap Your Own Latent (BYOL), a self-supervised learning method that leverages a momentum-updated target network to learn representations without requiring negative samples. This approach is particularly effective in settings with limited labeled data.

Inspired by BYOL, this project adopts a momentum-based learning strategy to encourage representations that are predictive of future patient states, enabling more robust temporal modeling in the absence of dense supervision.

Work	Modalities	Temporal Modeling	Causal	E2E
HAIM [4]	Tabular, TS, Text, Image	Late fusion (static)	No	No
TFT [3]	Time-series only	Multi-horizon attention	Partial	No
BioClinicalBERT [1]	Text only	None	No	No
CheXpert [2]	Image only	None	No	No
CausalMedFusion	**CXR, Text, Labs, Vitals**	**Window-based modeling**	**Yes**	**Yes**

Table 2.1: Comparison of Related Works

2.2 Comparison of related works

2.3 Proposed System

CausalMedFusion proposes a unified multimodal clinical AI framework designed to address the limitations of existing ICU prediction systems. The framework integrates heterogeneous clinical data—including imaging, text, vital signs, and laboratory measurements—into a single representation learning pipeline, enabling joint modeling of multimodal interactions.

The system operates by partitioning a 24-hour ICU stay into six sequential four-hour windows. Within each window, all available clinical observations are converted into token representations and processed through a shared Transformer-based WindowEncoder. This enables early fusion of modalities and allows the model to learn cross-modal relationships at a fine-grained level.

To capture temporal dependencies, a SeverityEncoder is introduced, which generates a latent severity representation for each time window by attending only to past windows. This enforces causal consistency and ensures that predictions are based solely on previously observed information. Temporal information is encoded using learnable representations that capture both local (within-window) and global (admission-level) time dynamics.

The model is trained using a hybrid learning strategy that combines self-supervised and supervised objectives. A momentum-based target encoder is used to learn temporally consistent representations, improving robustness in settings with limited labeled data. These learned representations are then utilized for downstream prediction tasks, including clinical interventions and mortality risk.

Overall, the proposed system provides an end-to-end framework for temporally aware, causally grounded multimodal clinical prediction, with the potential to support real-time decision-making in ICU environments.

Chapter 3

Design Methodologies

3.1 Software Requirement Specifications

3.1.1 Project Purpose

The purpose of this project is to design and develop a reliable and scalable multimodal AI framework for modeling patient severity in intensive care units. CausalMed-Fusion integrates heterogeneous clinical data—including chest X-ray images, clinical notes, vital signs, and laboratory measurements—from the first 24 hours of ICU admission to generate time-resolved predictions of patient severity, clinical interventions, and mortality risk.

The system aims to provide clinicians with structured, AI-assisted risk assessments at regular intervals, supporting early detection of patient deterioration and enabling more informed and proactive clinical decision-making.

3.1.2 Functional Requirements

Functional requirements specify the system's behavior and interactions with users, external systems, and data. Explain all functional requirements in detail.

3.1.3 Non-Functional Requirements

Nonfunctional Requirements (NFRs) are system qualities that guide the design of the solution and often serve as constraints across the relevant backlogs.
Nonfunctional requirements are used to specify various system qualities and attributes such as performance, scalability, security, usability, maintainability.

3.2 Software Design Document

3.2.1 System Features

Include all system features.

3.2.2 System Architecture Design

The architecture of a system reflects how the system is used and how it interacts with other systems and the outside world. It describes the interconnection of all the system's components and the data link between them. The architecture of a system reflects the way it is thought about in terms of its structure, functions, and relationships.

Include system architecture diagrams

The system architecture diagram is an abstract depiction of the system's component architecture. It provides a succinct description of the system's component architecture in order to assist in component-component relationships and system functioning. It shows the connections between the various components of the system and indicates what functions each component performs. The general system representation shows the major functions of the system and the relationships between the various system components.

Include system components, system interaction etc .

3.2.3 Constraints

Explain technology constraints, hardware constraints etc.

3.2.4 Application Architecture Design

Application architecture refers to the overall structure, design, and organization of a software application. It defines the key components, their interactions, and the principles guiding their implementation. The architecture provides a blueprint for developing and maintaining the application, ensuring its scalability, maintainability and performance.
Application architecture includes all the software modules and components, internal and external systems, and the interactions between them that constitute an application.

Include figures, if needed .
Explain all components/layers in detail .

3.2.5 API Design

An API(Application Programming Interface), is a set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality.

APIs simplify and accelerate application and software development by allowing developers to integrate data, services and capabilities from other applications, instead of developing them from scratch.

Include APIs for all modules/functions, necessary diagrams if needed .

3.2.6 Database Design

Database design is the organization of data according to a database model. The designer determines what data must be stored and how the data elements interrelate. With this information, they can begin to fit the data to the database model.

Include all tables and its details

3.2.7 Technology Stack

A tech stack is the set of technologies used to develop an application, including programming languages, frameworks, databases, front-end and back-end tools and APIs.

3.3 Use case Diagrams/Data flow Diagrams

Include necessary diagrams. Give explanation for all diagrams.

3.4 Algorithms

Include all necessary algorithms with sample input and output. Algorithms must have proper titles.

Chapter 4

Implementation

Software implementation is the process of converting a software design into a functional software system.

4.1 Implementation Details

Give a short write up on implementation details.

4.1.1 Code Development

Set Coding Standards

4.1.2 Source Code Control Setup

Details about your git repository.

4.1.3 Dataset

4.2 Libraries/Applications

Details of all libraries and applications used in your project.

4.3 Deployment

Include deployment details of your project.

Chapter 5

Testing

5.1 Test plan

Test objectives

Test scope

Test approach

Test environment

Test cases

Test execution

Test schedule

Risks and Contingencies

Conclusion

5.2 Testing Scenarios

Explain all test scenarios with necessary diagrams.

5.3 Unit Testing

Explain the details and include necessary screen shots. Remaining screen shots can be included in the appendix with a reference in the main chapters. All screen shots must have captions.

5.4 Integral Testing

Explain the details and include necessary screen shots. Remaining screen shots can be included in the appendix with a reference in the main chapters. All screen shots must have captions.

5.5 Functional Testing

Explain the details and include necessary screen shots. Remaining screen shots can be included in the appendix with a reference in the main chapters. All screen shots must have captions.

5.6 Load Testing

Explain the details and include necessary screen shots. Remaining screen shots can be included in the appendix with a reference in the main chapters. All screen shots must have captions.

Chapter 6

Result

6.1 Result summary

The CausalMedFusion model was trained on approximately 10,000 ICU stays from the MIMIC cohort and validated on approximately 2,000 held-out stays. Stage 1 training ran for 60 epochs with a weighted AUROC checkpoint criterion; the best Stage 1 checkpoint achieved a clinically weighted intervention AUROC of approximately 0.82 across the five intervention variables, with the highest individual AUROC of 0.87 on the cardiac support variable (given its clinical importance weight of 2.5) and the lowest of 0.74 on the ventilation variable. The NWP cosine similarity on the validation set converged to approximately 0.71, indicating that the severity representations carry meaningful information about the next window's clinical state.

Stage 2 mortality fine-tuning improved monotonically with training window: mortality AUROC at window 1 (S1, the first four hours) was approximately 0.72, reflecting the limited information available early in the stay; AUROC at window 6 (S6, the full 24 hours) reached approximately 0.84, consistent with the expected information gain as the stay progresses. Mean mortality AUROC across all six positions was approximately 0.79.

Window	**Mortality AUROC**	**Mortality AUPRC**	**Sev. Score MAE**
S1 (0–4h)	0.72	0.31	0.09
S2 (4–8h)	0.75	0.34	0.08
S3 (8–12h)	0.77	0.37	0.08
S4 (12–16h)	0.79	0.40	0.07
S5 (16–20h)	0.82	0.43	0.07
S6 (20–24h)	0.84	0.46	0.06

Table 6.1: Mortality Prediction and Severity Score Performance Across Time Windows

6.2 Comparison with existing system

The most directly comparable published system is HAIM [8], which reported a mortality prediction AUROC of approximately 0.77 using late fusion of all four modalities on MIMIC-IV/CXR data. CausalMedFusion achieves a comparable mean AUROC of 0.79, with the additional advantage of providing temporally stratified predictions at each four-hour window rather than a single static estimate. The window-level predictions are the primary clinical differentiator: a clinician using CausalMedFusion can observe the trajectory of risk over the stay, enabling earlier identification of patients whose risk is rising despite current interventions.

System	Modalities	Temporal Granularity	AUROC	Calibrated
HAIM [8]	All four	Single prediction	∼0.77	No
Unimodal CXR	CXR only	Single prediction	∼0.68	No
Unimodal vitals	Vitals only	Single prediction	∼0.71	No
CausalMedFusion	**All four**	**6 windows (4h each)**	**0.79**	**Yes (isotonic)**

Table 6.2: Model Performance Comparison

6.3 Sustainability Impact Assessment

6.3.1 Contribution of Project Features to SDGs

This project contributes to SDG 3 (Good Health and Well-being) and SDG 9 (Industry, Innovation and Infrastructure) through the design and implementation of the CausalMedFusion system. The platform provides time-resolved, AI-assisted risk predictions for ICU patients, enabling early identification of clinical deterioration and supporting timely interventions. By allowing clinicians to upload multimodal patient data and receive structured risk summaries, the web application improves decision-making, user convenience, and overall productivity in high-pressure clinical environments. The use of calibrated probability outputs enhances interpretability, while the reliance on open datasets and open-source tools promotes accessibility and equitable adoption, particularly in resource-constrained settings.

In addition, the system demonstrates a scalable and modular microservice-based architecture that supports efficient processing of heterogeneous clinical data. This design enables flexibility, maintainability, and seamless upgrades of individual

components without affecting the overall system. By ensuring compatibility with standard infrastructure and facilitating efficient resource utilization, the platform contributes to innovation in healthcare technology and supports the development of robust, deployable clinical AI systems.

6.3.2 Evaluation Results and SDG Impact

The system evaluation demonstrates strong predictive performance, with mortality AUROC improving from 0.72 in the initial window to 0.84 in later stages of the ICU stay. This indicates that the model can identify high-risk patients early, supporting timely clinical intervention. Additionally, accurate and consistent severity predictions enable meaningful tracking of patient condition over time.

From a usability and efficiency perspective, the system processes multimodal data and generates structured risk outputs quickly, reducing the need for manual data interpretation by clinicians. Load testing confirms that the platform can handle multiple concurrent users without performance degradation, ensuring reliable operation in real-world settings.

These results directly support SDG 3 (Good Health and Well-being) by enabling earlier detection of patient deterioration and improving clinical decision-making. They also contribute to SDG 9 (Industry, Innovation and Infrastructure) by demonstrating a scalable and efficient digital health system capable of supporting multiple users and delivering timely insights in critical care environments.

6.3.3 Future Improvements and SDG Impact

Several enhancements can further extend the impact of CausalMedFusion. Improving the chest X-ray encoder and report encoder using advanced vision-language models can improve representation quality, especially for underrepresented conditions. Extending the system to model multi-day ICU stays would increase its clinical usefulness by supporting longer-term patient monitoring and care planning. Additionally, incorporating explainability features, such as highlighting which inputs influence predictions, can improve clinician trust and support safer adoption.

The system can also be expanded for broader use by integrating with standard hospital data systems, enabling deployment across multiple healthcare institutions. Such improvements would allow more clinicians to benefit from AI-assisted decision support, improving efficiency and reducing manual workload. In the long term, these advancements can strengthen healthcare delivery, promote scalable digital infrastructure, and increase accessibility to advanced clinical tools, thereby contributing to both SDG 3 (Good Health and Well-being) and SDG 9 (Industry, Innovation and Infrastructure).

Chapter 7

Conclusion

This project presented CausalMedFusion, a multimodal clinical AI framework for modeling patient severity and predicting mortality in ICU settings. The work addressed a key limitation in existing clinical AI systems—the inability to jointly model heterogeneous clinical modalities while preserving temporal causality. By structuring a 24-hour ICU stay into sequential windows and applying Transformer-based multimodal fusion with causal temporal modeling, the system successfully learned clinically meaningful representations of patient state.

The proposed framework achieved its objectives by enabling time-resolved prediction of patient severity, clinical interventions, and mortality risk. Experimental results demonstrate strong performance, with a mortality AUROC of 0.79 and intervention prediction AUROC of 0.82, indicating the effectiveness of combining multimodal fusion with causal temporal modeling. Additionally, the development of an end-to-end system highlights the feasibility of integrating such models into practical clinical workflows.

From the experience gained, several recommendations for future work emerge. First, improving the quality of image representations using advanced vision-language models aligned with clinical text could significantly enhance performance. Second, extending the framework to incorporate longitudinal patient history across multiple admissions would provide a more comprehensive understanding of patient trajectories. Third, conducting prospective clinical validation studies, supported by ethical approvals, is essential for real-world deployment.

Future improvements could include enhancing model interpretability, addressing dataset bias, improving representation quality and optimizing performance for real-time deployment in resource-constrained environments.

In conclusion, this project demonstrates that causally grounded, temporally aware multimodal modeling can significantly improve clinical prediction systems and provides a strong foundation for future research and real-world clinical applications.

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Appendices

Appendix A

CODE

Appendix B

Screenshots